

BESS RFP Appendix 1

Technical Specification – BESS Main Equipment
Requirements

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1. ABBREVIATIONS, DEFINITIONS AND INTERPRETATION

1.1. Abbreviations

AC	Alternate Current
ANSI	American National Standards Institute
BMS	Batteries Management System
BOM	Bill of Material
CCTV	Closed Circuit Television
CDF	Constructional Data Form
CFC	Chlorofluorocarbons
CFD	Computational Fluid Dynamics
DC	Direct Current
DMZ	Demilitarized Zone
DSP	Digital Signal Processor
EMS	Energy Management System
FRT	Fault Ride Through
HMI	Human Machine Interface
HVAC	Heating, Ventilation and Air Conditioning
HVRT	High Voltage Ride Through
ICS	Industrial Control systems
IEC	International Electrotechnical Commission
IED	Intelligent Electronic Device
IEEE	Institute of Electrical and Electronics Engineers
IGBT	Insulated Gate Bipolar Transistor
ILAC	International Laboratory Accreditation Cooperation
IP	Ingress Protection
ISO	International Organization for Standardization
IT	Information Technology
KPI	Key Performance Indicator
LDC	Load Dispatch Center
LED	Light Emitting Diode
LFP	Lithium Ferro Phosphate
LOTO	Lockout / Tagout
LV	Low Voltage
LVRT	Low Voltage Ride Through
MTBF	Mean Time Between Failures
MTTF	Mean Time To Failure
MTTR	Mean Time To Repair
MV	Medium Voltage
N/A	Not Applicable
NIST	National Institute of Standards and Technology
OEM	Original Equipment Manufacturer
OT	Operational Technology
O&M	Operation and Maintenance

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PCS	Power Conversion System
PID	Proportional-Integral-Derivative
PPE	Personal Protective Equipment
PSCAD	Power Systems Computer Aided Design
PSSE	Power System Simulator of Engineering
RAM	Reliability, Availability and Maintainability
RMU	Ring Main Unit
SCADA	Supervisory Control and Data Acquisition
SCMS	Substation Control and Monitoring System
SIL	Safety Integrity Level
SL-T	Security Level Target
SoC	State of Charge
SoH	Stage of Health
TCP	Transmission Control Protocol
UL	Underwriters Laboratories
UPS	Uninterruptible Power Supply
VFD	Variable Frequency Drive
VSG	Virtual Synchronous Generator

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1.2. Interpretation

The Equipment shall comply with the requirements set out in this specification, any of its attachments, applicable local and international codes and standards and good industry practices. In case of any discrepancy, ambiguity or divergence between the requirements set out in the aforementioned documents, the most stringent requirement shall apply. Should there be any difference of opinion about the interpretation of requirements, the Buyer shall instruct the Seller as to the requirements to be implemented and the Seller shall, at its own cost, comply with that instruction.

2. BATTERY ENCLOSURE

2.1. Scope of Supply

Battery Enclosures shall integrate:

Table 1: Scope of Supply for Battery Enclosures

ID	Item	Scope	Remarks
1.01	Battery enclosure	[●]	
1.02	Battery cells	[●]	
1.03	Battery packs	[●]	
1.04	Battery racks	[●]	
1.05	Battery pack DC/DC converters	[●]	
1.06	Battery rack DC/DC converters	[●]	
1.07	String PCS	[●]	
1.08	DC panels	[●]	
1.09	LV AC panels	[●]	
1.10	Auxiliary panels	[●]	
1.11	Thermal management system	[●]	
1.11.1	HVAC	[●]	
1.11.2	Air temperature sensors	[●]	
1.11.3	Air relative humidity sensors	[●]	
1.11.4	Liquid cooling / heating system	[●]	
1.11.5	Forced air cooling / heating system	[●]	
1.12	Fire detection system	[●]	
1.12.1	Smoke detectors	[●]	
1.12.2	Heat detectors	[●]	
1.12.3	Flammable gas detectors	[●]	
1.12.4	Exhaust fan	[●]	
1.12.5	Deflagration venting panel	[●]	
1.12.6	Emergency push button	[●]	
1.12.7	Fire and gas alarms, audible and visual	[●]	
1.12.8	Control panel	[●]	
1.13	Fire suppression system	[●]	

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ID	Item	Scope	Remarks
1.13.1	Gas based agent	[●]	
1.13.2	Aerosol based agent	[●]	
1.13.3	Sprinklers	[●]	
1.13.4	Dry pipes	[●]	
1.14	BMS	[●]	
1.15	UPS	[●]	
1.16	Signal cables, power cables and other accessories for installation and operation	[●]	

2.2. General Requirements

Battery Enclosures shall:

- Be offered as a modular, fully integrated, prefabricated, tested containerized solution.
- Be newly manufactured. No used, reconditioned or salvaged equipment or material shall be allowed.
- Be of proven design and include components of proven technology with sufficient operational history.
- Be manufactured in manufacturing facilities certified according to ISO 9001 (Quality Management System) and ISO 14001 (Environmental Management System) by an internationally recognized certification authority.
- Deliver rated DC power capacity at 40 °C ambient temperature within the complete SoC range (0% to 100%).
- Be designed and constructed in full compliance with the requirements of UL 9540 and IEC 62619.
- Be designed and constructed considering the battery related hazards mentioned in IEC 61427-2.
- Be designed considering hazardous classification requirements as per NFPA 704.
- Be constructed of non-combustible material and have a minimum ingress protection rating of IP55.
- Have a minimum 60/60/60 fire resistance limit, all 6 faces, doors and deflagration venting panel included (meaning that fire resistance is 60 min for structural adequacy, 60 min for integrity and 60 min for insulation).
- Be protected from atmospheric corrosion and sand abrasion by means of protective paint in accordance with ISO 12944, the site corrosion category, as per ISO 9223, and the protective paint durability, as per ISO 12944. The protective paint shall be of a light ISO color to minimize heat absorption from solar radiation.
- Have weather resistant labelling on the external face of the doors as per ANSI Z535 standards and include type of battery technology, "Battery Energy Storage System" with a symbol in lightning bolt in triangle and emergency contact information.
- Have a non-walk in design. Operation and maintenance activities shall not require entering the Battery Enclosure.
- Be intrinsically safe, i.e., a foreseeable misuse such as an external short circuit may not lead to damage to the battery cells or create safety hazards, and include safety mechanisms of the battery on cell, pack, rack and battery system level. Passive functions (e.g., fuses and vents) and active functions (e.g., BMS commands) shall be specified.
- Include a monitoring and control system display for local monitoring and operation by O&M personnel. Alternatively, connection port for external notebook (laptop) is acceptable.

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- Include a drain syphon at the bottom of the enclosure with an adequate slope to ensure that, in case of any leakage by the thermal management system, all the leaked liquid is stored in the designed space. Alternative solution with drain valves, drain chute and drain connection point is acceptable.
- Include electrical protection devices, lock-in-tag-out (LOTO) system and emergency lighting.
- Integrate power, instrumentation, control and data cables in accordance with UL 2196.
- Integrate lighting and sockets for operation and maintenance activities.
- Allow the installation and removal of battery packs using appropriate lifting equipment.
- Have its noise emissions limited to 75 dB(A) at 1 m distance when operated at maximum electrical and thermal loads and under any weather condition.

The supplier of the Battery Enclosures shall recommend the foundations of the Battery Enclosures in accordance with the Site climatic conditions, the soil conditions and the seismic loads of the Site, retaining maximum wind speeds recorded in the region and thermal loads caused by expected fluctuations of materials and ambient temperatures.

2.3. Battery Cells

The battery cells shall:

- Have the technology and chemistry restricted to LFP.
- Be of the same manufacturer, family product, type, technology, model, series and capacity.
- Be manufactured with the same materials (brand and model) and, where possible, in the same factory.
- Be structurally identical to the ones tested and approved as part of the third-party certificates.
- Have insulated terminals. No live parts shall be exposed during maintenance and replacement activities.
- Be designed with particular attention to ease of isolation for maintenance and replacement activities.

2.4. Battery Packs

The battery packs shall:

- Have a minimum ingress protection rating of IP67.
- Be of the same manufacturer, family product, type, technology, model, series and capacity.
- Be manufactured with the same materials (brand and model) and, where possible, in the same factory.
- Be structurally identical to the ones tested and approved as part of the third-party certificates.
- Terminals shall be insulated. No live parts shall be exposed during maintenance and replacement activities.
- Include overcurrent protection device and any other required electrical protection devices.

The battery packs replacement, recycling and disposal protocol shall include:

- A list of steps from replacement on site until arrival at the recycling facility.
- A description of the recycling process.
- A list of waste materials generated in the recycling process.
- A tracking system to follow the progress of the recycled quantities.
- A recyclability certificate of the battery packs by a qualified third-party.
- An assessment on how the proposed protocol complies with the Basel Convention.

2.5. Battery Racks

The battery racks shall:

- Integrate number of battery packs connected in series that maximizes the battery rack voltage, while keeping it within the PCS voltage limits, so the DC cables losses and cross section is minimized
- Ensure a 100 mm clearance from the Battery Enclosure floor to the battery packs.
- Ensure uniform spacing between battery packs.

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- Be leveled and fixed to the Battery Enclosure.
- Be made of galvanized or painted steel.

2.6. DC/DC Converters

DC/DC converters shall:

- Enable bidirectional power transfer between battery pack and battery rack or battery rack and DC panel.
- Have a minimum ingress protection rating of IP54 and comply with IEC 62477-1 and IEEE 1547.
- Have an operating voltage range able to accommodate the battery pack or battery rack voltage.
- Include undervoltage, overvoltage, overcurrent, overheating and insulation protection.
- Support for remote and local control and be real time monitored in the BMS.
- Include emergency shutdown provisions in case of critical faults.

2.7. String PCS

The PCS shall integrate:

- Power electrics module:
 - Solid state electronic devices, including IGBT.
- DC termination module with:
 - Bus terminals for DC cables connection.
 - Fuses.
 - Overcurrent protection device.
 - Overvoltage protection device.
 - Surge protection device.
 - Isolation switch.
 - Insulation monitoring device.
 - Any other required electrical protection devices.
 - Electrical protection devices shall be installed on each polarity.
- AC termination module with:
 - Bus terminals for LV cables connection.
 - Overcurrent protection device.
 - Overvoltage protection device.
 - Reverse current protection (isolation) device.
 - Surge protection device.
 - Microcontroller based automatic circuit breakers.
 - Insulation monitoring device.
 - Any other required electrical protection devices.
- Control module with:
 - External notebook (laptop) connection port for local monitoring and operation by O&M personnel.
 - Master control board.
 - Power supply unit.
 - Fieldbus communications unit.
 - Control panel including start/stop button, LED indicators for PCS status (operation (charge, discharge and idle) or stop (faults)) and emergency push button.
- Its own cooling system.
- Temperature control and moisture content control.
- Doors interlock system preventing the doors from being opened while energized.

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The PCS shall be able to perform the following electrical protection functions, which triggering shall imply the power electronics tripping and the opening of the DC and AC circuits to isolate the PCS:

- Undervoltage and overvoltage protection, DC and AC.
- Overcurrent protection, DC and AC.
- Underfrequency and over frequency protection.
- Surge protection, DC and AC.
- Ground fault protection.
- Overheating protection.
- Smoke detection, alarm and trip.
- Insulation protection.

The PCS shall:

- Be bi-directional and suitable to provide quick responses and required reactive power.
- Be of the same manufacturer, family product, type, technology, model, series and capacity.
- Be manufactured with the same materials (brand and model) and, where possible, in the same factory.
- Be structurally identical to the ones tested and approved as part of the third-party certificates.
- Be of proven design and include proven IGBT technology with sufficient operational history.
- Deliver its rated power capacity at 40 °C ambient temperature within the complete SoC range (0% to 100%).
- Be able to generate reactive power from AC power / energy without consuming DC power / energy.
- Include self-protective and self-diagnostic features to protect itself from damage in the event of component failure or from parameters beyond safe range. The self-protective features shall not allow the PCS to be operated in a manner that may be unsafe or damaging.
- Be programable and alterable only by personnel with appropriate clearance and access.
- Have its noise emissions limited to 75 dB(A) at 1 m distance when operated at maximum electrical and thermal loads and under any weather condition.

The PCS shall have:

- Minimum ingress protection rating of IP21 for indoor installation.
- Minimum ingress protection rating of IP65 for outdoor installation.
- Provisions to prevent moisture condensation.
- Provisions to prevent the entrance of rodents, insects or any materials into air intake and exhaust ports.
- A DC voltage range wider than the DC voltage range of the battery racks.
- A maximum DC current greater than the DC current demanded by the connected battery racks during charge and discharge operating at maximum load.
- Charge and discharge efficiency greater than 98.6% at nominal load.
- Charge and discharge efficiency greater than 98.3% at any load point.
- Output current total harmonic distortion lower than 3% at rated power output.
- Output current individual harmonics lower than 1% at rated power output.

The PCS shall provide:

- Active and reactive power control.
- Grid supporting (grid following) functionalities:
 - LVRT / HVRT.
 - $\pm 100\%$ PQ.
- Grid controlling (grid forming) functionalities:
 - Voltage control.

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- Primary and secondary frequency control.
- Ramp rate control.
- Virtual inertia (VSG mode).
- High short circuit reactive current contribution capability during FRT conditions.
- Local and remote monitoring and control functionalities.

2.8. DC Panels

The DC panels shall integrate:

- Copper DC busbars.
- Fuses.
- Overcurrent protection device.
- Overvoltage protection device.
- Surge protection device.
- BMS operated DC contactors.
- Any other required electrical protection devices.

The DC panel shall interface the battery racks with the DC terminal of the PCS for protection and isolation requirements at Battery Enclosure level. The number of battery racks connected to each DC panel shall be appropriately designed according to the protection system including in terms of fuse nominal and short circuit current rating. A proper protection coordination study shall be carried out between battery rack, DC panel and PCS before finalizing the DC protection design.

2.9. LV AC Panels

The LV AC panels shall integrate:

- Copper AC busbars.
- Overcurrent protection device.
- Overvoltage protection device.
- Reverse current protection (isolation) device.
- Surge protection device.
- Microcontroller based automatic circuit breakers.
- Insulation monitoring device.
- Any other required electrical protection devices.

The LV AC panel shall interface the AC terminal of the PCS with the MV Station. The number of PCS connected to each LV AC panel shall be appropriately designed according to the protection system including in terms of fuse nominal and short circuit current rating. A proper protection coordination study shall be carried out LV AC panel, PCS and MV Station before finalizing the LV AC protection design.

2.10. Auxiliary Panels

The auxiliary loads within the Battery Enclosures shall not be self-supplied from the battery racks located within the same Battery Enclosure. Each Battery Enclosure shall be supplied from the auxiliary transformers of the BESS Plant.

The auxiliary panel shall allow supply of power to the thermal management system and to other devices of the Battery Enclosure through an auxiliary external power supply. The auxiliary panels shall integrate:

- Auxiliary power supply AC cable terminals.
- Overcurrent protection device.
- Surge protection device.
- Circuit breakers.
- Any other required electrical protection devices.

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2.11. Thermal Management System

Thermal management system shall integrate:

- HVAC, in accordance with NFPA 90A, along with air temperature and relative humidity sensors to regulate the air conditions into which electrical and electronic components operate.
- Liquid cooling / heating system to maintain the battery cells temperature within the permissible operating temperature range under the performance warranty. For the avoidance of doubt, forced air cooling / heating system is not allowed.

2.11.1. General Requirements

HVAC and liquid cooling / heating system design shall:

- Permit the Battery Enclosures to deliver rated DC power capacity at 40 °C ambient temperature within the complete SoC range (0% to 100%).
- Consider site's solar radiation, site's ambient temperature and the heat emitted by the systems and components inside the Battery Enclosures operating at full load as thermal input to the CFD analyses required for their sizing.
- Have a cooling capacity equal to 125% of the maximum cooling load to tackle with the potential cooling load demand increase derived from the battery cells aging.
- Link their operation to the fire detection system switching them to an appropriate operating mode in case of fire detection.
- Include self-cleaning air filters with automated systems that periodically remove dust, sand and contaminants without the need of manual cleaning such as pulse jet cleaning, mechanical shaking and electrostatic precipitators.
- Include VFDs to minimize noise emissions.
- Include wall boxes and collars to seal passings through compartments within the Battery Enclosure.
- Include CFC free refrigerants.

2.11.2. Liquid Cooling / Heating System

The liquid cooling / heating system refrigerant circuit shall integrate:

- Refrigerant.
- Compressor, equipped with VFD.
- Evaporator to interface with the coolant circuit.
- Heat exchanger.
- Electronic expansion valve to control the flowrate of the refrigerant circuit.
- Temperature, pressure and flow meters linked with the BMS and automatically detecting refrigerant leakage.
- Drain points, pots and vessels incorporated with anti-leakage plugs.
- Air vents and exhaust valves.
- Piping, valves and fittings which in all cases shall be airtight and include double lock.

The liquid cooling / heating system coolant circuit shall integrate:

- Coolant, colored for ease of detection of leakages. Organic acid coolants are recommended.
- Heater, dimensioned to ensure battery cells operate within an acceptable temperature range.
- Pumps, equipped with VFD.
- Temperature, pressure and flow meters linked with the BMS and automatically detecting refrigerant leakage.
- Drain points, pots and vessels incorporated with anti-leakage plugs.

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- Air vents and exhaust valves.
- Piping, valves and fittings which in all cases shall be airtight and include double lock.
- Coolant replenishing circuit which shall incorporate a filling/drain port for coolant replenishment. The port shall be blocked by an anti-leakage plug. The replenishing circuit shall include the required storage and expansion tanks, pumps and valves and shall allow for automatic coolant refilling from the tank. When the coolant level in the tank is below the minimum level, a liquid level sensor shall send an alarm to refill the tank.

The liquid cooling / heating system design shall:

- Have provisions to work as a heating circuit in case of low ambient temperature.
- Prevent condensation within the Battery Enclosures. Any ductwork, casing, piping and equipment with surface temperatures below the ambient dew point temperature shall be insulated.
- Grant access to equipment, valves, drains and air vents.
- Minimize low and high points. Adequate high- and low-pressure alarms shall be included in the system.
- Ensure that the refrigerant and the coolant circuits piping do not run in trays with electrical cables.
- Ensure, through pressure sensors, that system pressure is within the limits at all times. An inlet pressure lower than the established minimum acceptable value shall automatically trigger the activation of the coolant replenishing circuit.
- Be in such way that each coolant subcircuit corresponding to each row of battery racks is equipped with valves and drain port with anti-leakage plug so that, if a battery rack needs to be replaced, only one coolant subcircuit needs to be opened.

2.12. Fire Detection System

The fire detection system shall integrate:

- Smoke detectors with a detection range of -20/+75 °C, in accordance with NFPA 72.
- Heat detectors with a detection range of -20/+75 °C, in accordance with NFPA 72.
- Flammable gas detectors (H₂ and CO) with a detection range of -20/+75 °C.
- Exhaust fan. The exhaust fan shall be electrically classified Class I Div 1 and shall be designed to withstand the high temperatures expected from the battery off-gas during thermal runaway.
- Deflagration venting panel. The deflagration venting panel, in accordance with NFPA 68, shall be installed in the Battery Enclosure ceiling ensuring the forces resulting from the deflagration are released vertically, not affecting O&M personnel nor surrounding components.
- Fire resistant and flame retardant power and control cables, in accordance with UL 2196.
- Emergency push button installed externally. The emergency push button shall trigger the fire alarm and the de-energization of the related MV Station.
- Fire and gas alarms, audible and visual. The audible alarms shall integrate a buzzer. The visual alarms shall integrate a flashlight installed externally to the Battery Enclosure.
- Control panel, in accordance with UL 864, along with control logics:
 - The flammable gas ventilation logic, in accordance with NFPA 69, and in conjunction with the flammable gas detectors and the exhaust fan, shall grant a flammable gas concentration below 25% of its lower flammability limit (12.5% v/v for CO and 4% v/v for H₂) at all times.
 - In case of simultaneous detection of smoke and gas, the gas ventilation shall be prioritized over the fire suppression system activation.
 - Voting logics involving detectors of the same and different type shall be implemented to avoid inadvertent alarms.

2.13. Fire Suppression System

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The fire suppression system shall:

- Be automatically or manually activated by the fire detection system.
- Be capable of addressing various fire hazards, including electrical fires (non-thermal runaway) and thermal runaway events, as well as incorporating measures for explosion mitigation or prevention.
- Integrate clean agent or aerosols as fire suppression agent as first level barrier against small electrical fires.
- Be designed in accordance with NFPA 2001 for clean agent-based systems and with NFPA 2010 for aerosol-based systems.
- Integrate galvanized steel pipes for clean agent-based systems.

2.14. BMS

The BMS shall:

- Balance battery packs SoC, actively or passively.
- Perform battery packs SoC recalibration.
- Manage the thermal management system.
- Manage the fire detection system.
- Manage the UPS.
- Manage the PCS to set operational conditions (current, voltage and power) within the limits of the battery packs and racks, both during charge and discharge.
- Perform electrical safety functions:
 - Undervoltage, overvoltage and voltage imbalance alarm and protection for battery packs and racks.
 - Overcurrent and current imbalance alarm and protection for battery packs and racks.
 - Overcharge, overdischarge and charge imbalance alarm and protection for battery packs and racks.
 - Temperature default, excess and imbalance alarm and protection for battery packs and racks.
 - Disconnection of battery racks exhibiting abnormal operating conditions.
 - Report any malfunctioning device based on temperature, voltage, and SoC real-time analysis.
 - Diagnosis and management of faults.
- Measure the battery packs SoC with a relative error not greater than $\pm 3\%$.
- Measure the battery packs temperature with an absolute error not greater than $\pm 1^\circ\text{C}$.
- Measure the battery packs and racks voltage with an absolute error not greater than $\pm 10\text{ mV}$.
- Measure the battery packs and racks current with a relative error not greater than $\pm 0.5\%$.
- Include self-diagnosis, self-testing and self-restoration capabilities.
- Include back-up and reset system to enable data loss management.
- Have a minimum communication speed capability of at least 10 Gbps.
- Communicate with NTP protocol to achieve time stamp synchronization.
- Include software free of virus and malware and lifetime license for such software with no cost for:
 - new versions of the software, as the same may become available;
 - updates consisting of or including modifications or improvements and routine bug fixes to the existing features of the software including specific fixes to bugs whether notified, reported by the user or included in the new versions of the software; and
 - support by e-mail, in respect of faults, failures or other issues with the installation or proper functioning of the software.
- Interoperate with the rest of the monitoring and control systems in standardized approved protocols.
- Include authentication and authorization features to avoid unauthorized access and control.

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The BMS shall monitor, at least, the following data:

Table 1: Data monitored by BMS

Equipment / System	Data	Remarks
Battery Cell	DC current	Calculated as the battery rack DC current divided by the number of parallel circuits in the battery pack
	DC voltage	Measured
	Temperature	Measured
Battery Pack	DC current	Calculated as the battery rack DC current
	DC voltage	Calculated as the sum of the battery cells DC voltage connected in series in the battery pack
	Temperature	Reported as the average, minimum and maximum temperature of the cells in the battery pack
Battery Rack	DC current	Measured
	DC voltage	Measured
	Temperature	Reported as the average, minimum and maximum temperature of the cells in the battery rack
	SoC	Calculated
	SoH	Calculated
Battery Enclosure	DC current	Calculated as the sum of the battery racks DC current
	DC voltage	Calculated as the minimum of the battery racks DC voltage
	Temperature	Reported as the average, minimum and maximum temperature of the cells in the Battery Enclosure
	SoC	Calculated
	SoH	Calculated
	Charge current limit	Calculated
	Charging power limit	Calculated
	Discharge current limit	Calculated
	Discharging power limit	Calculated
	Available input energy	Calculated
	Available output energy	Calculated
	Total operating time since first use	Measured in hours
	Total number of cycles since first use	Calculated
	Total energy delivered since first use	Measured in MWh
	Total energy delivered since last charge	Measured in MWh
	Hard-wired emergency off-input	-
	Hard-wired emergency off-output	-
	Status: charge/discharge	-
String PCS	Operational data (electrical parameters)	Measured
	DC current	Measured
	DC voltage	Measured
	AC current	Measured

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Equipment / System	Data	Remarks
	AC voltage	Measured
	Apparent power	Measured
	Active power	Measured
	Reactive power	Measured
	Power factor	Calculated
	Frequency	Measured
	Operating mode / Workstate	-
	Status and alarms	-
Auxiliary Panel	Current	Measured
	Voltage	Measured
	Power	Measured
	Status and alarms	-
Thermal Management System (HVAC)	Air temperature	Measured
	Air relative humidity	Measured
	Power consumption	Measured
	Operating mode / Workstate	-
	Status and alarms	-
Thermal Management System (Liquid cooling / heating system)	Refrigerant temperature, pressure and flow	Measured
	Coolant temperature, pressure and flow	Measured
	Power consumption	Measured
	Operating mode / Workstate	-
	Status and alarms	-
Fire Detection System	Smoke detectors status and alarms	-
	Heat detectors status and alarms	-
	Flammable gas detectors status and alarms	-
	Exhaust fan status and alarms	-
	Battery Enclosure fire alarms	-
Fire Suppression System	Status and alarms	-
BMS Controller	Devices information	-
	Operator parameters and permission	-
	Status and alarms	-
UPS	Current	-
	Voltage	-
	SoC	-
	Status and alarms	-
Alarms	General alarms and failures	-
User logins	Type, local or remote	-
	Time and date	-
	Parameters update	-
	Commands	-

2.15. UPS

The UPS shall:

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- Have a power and an energy capacity sized for 2 h backup of essential DC and AC consumers to safely shutdown and monitor the Battery Enclosure in case of loss of auxiliary power supply:
 - BMS.
 - Fire detection system.
 - Fire suppression system.
 - Electrical protection devices.
 - Emergency lighting.
- Be able to deliver its rated output under the Site climatic conditions.
- Provide the short-circuit current necessary to ensure a selective trip of downstream electrical protection devices.
- Maintain the output voltage within $\pm 1\%$ of the rated output voltage under the following conditions:
 - $\pm 5\%$ of the rated input frequency.
 - $\pm 10\%$ of the rated input voltage.
 - $\pm 100\%$ of the rated power output.
- Be installed indoor and shall be cooled by natural or forced air ventilation.

2.16. Logo Imprint

Each Battery Enclosure must have Masdar’s logo imprinted on the side of the Battery Enclosure and Masdar’s “Infinitor” logo imprinted on the center of the top side of the Battery Enclosure for aerial view. No other logos shall be imprinted on the Battery Enclosures. The logos’ color value shall be C100 M22 Y0 K29. Samples of the logos can be found below:

Table 2: Samples of Masdar’s logos

Masdar’s Infinitor Logo	Masdar’s Infinitor Logo
	

The logo imprint must be durable and resistant to wear, fading, and damage under Site climatic conditions, including exposure to radiation, temperature fluctuations, humidity, and chemicals. The logo imprint must align with Masdar’s quality benchmarks for visibility and accuracy. The logos shall be in the designated positions and sizes as per Masdar’s branding guidelines.

API and EPS files and Masdar’s branding guidelines will be shared before the commencement of production.

2.17. Certification

The Battery Enclosures shall have valid certifications issued by reputable ILAC-accredited testing institutions. The samples tested for such certifications must be representative of the production units. Full test reports, including the CDF and any other relevant annexes, shall be provided along with the certificates. The CDF serves as the list of components used during the certification testing of a Battery Enclosure.

To ensure that the Battery Enclosure is certified, the components in its BOM must be covered in the CDF of the required certificates and test reports. If the BOM of the Battery Enclosure does not match the CDF, the corresponding certification testing shall be repeated to certify the BOM of the Battery Enclosures.

The certifications required for the Battery Enclosure are for the following standards in their latest editions:

Table 3: Battery Enclosure Certificates

Component	Data	Remarks
Battery Cell	IEC 62281	Safety of primary and Secondary Lithium Cells and Batteries During Transport.
	IEC 62485-5	Safety Requirements for Secondary Batteries and Battery Installations - Part 5: Safe Operation of Stationary Lithium-Ion Batteries.

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Component	Data	Remarks
	IEC 62619	Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Secondary Lithium Cells and Batteries for Use in Industrial Applications.
	IEC 62620	Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Secondary Lithium Cells and Batteries for Use in Industrial Applications.
	UL 1642	Lithium Batteries.
	UL 1973	Batteries for Use in Stationary, Vehicle Auxiliary Power and Light Electric Rail (LER) Applications.
	UL 9540A	Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems.
	UN 38.3	Transport of Dangerous Goods – Lithium Batteries.
Battery Pack	IEC 60529	Degrees of protection provided by enclosures (IP Code).
	IEC 62281	Safety of primary and Secondary Lithium Cells and Batteries During Transport.
	IEC 62619	Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Secondary Lithium Cells and Batteries for Use in Industrial Applications.
	UL 1973	Batteries for Use in Stationary, Vehicle Auxiliary Power and Light Electric Rail (LER) Applications.
	UL 9540A	Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems.
	UN 38.3	Transport of Dangerous Goods – Lithium Batteries.
Battery Rack	UL 1973	Batteries for Use in Stationary, Vehicle Auxiliary Power and Light Electric Rail (LER) Applications.
	UL 9540A	Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems.
Battery Enclosure	EN 12094-1	Fixed Firefighting Systems. Components for Gas Extinguishing Systems.
	EN 54-2	Fire Detection and Fire Alarm Systems. Control and Indicating Equipment.
	EN 54-4	Fire Detection and Fire Alarm System.
	EN 1363-1	Fire Resistance Tests - Part 1: General Requirements.
	EN 1363-2	Fire Resistance Tests - Part 2: Alternative and Additional Procedures.
	IEC 60529	Degrees of protection provided by enclosures (IP Code)
	IEC 61427-2	Secondary Cells and Batteries for Renewable Energy Storage – General Requirements and Methods of Test – Part 2: On-Grid Applications.
	IEC 62485-5	Safety Requirements for Secondary Batteries and Battery Installations - Part 5: Safe Operation of Stationary Lithium-Ion Batteries.
	IEC 62933-5-2	Electrical Energy Storage (EES) Systems - Part 5-2: Safety Requirements for Grid-Integrated EES Systems - Electrochemical-Based Systems.
	IEC 63056	Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Secondary Lithium Cells and Batteries for use in Electrical Energy Storage Systems.
	NFPA 68	Explosion Protection by Deflagration Venting.
	NFPA 69	Explosion Prevention Systems.
	NFPA 72	National Fire Alarm and Signaling Code.
	UL 855	Installation of Stationary Energy Storage Systems.
	UL 1973	Batteries for Use in Stationary, Vehicle Auxiliary Power and Light Electric Rail (LER) Applications.

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Component	Data	Remarks
	UL 9540	Energy Storage System Requirements.
	UN 3536	Lithium Batteries Installed in Cargo Transport Unit – Lithium-Ion Batteries or Lithium Metal Batteries.
	UN 38.3	Transport of Dangerous Goods – Lithium Batteries.
String PCS	Grid code	Electricity Transmission Code as per the Malaysia Grid Code
	IEC 60068-2-30	Environmental testing - Part 2-30: Tests - Test Db: Damp heat, cyclic (12 h + 12 h cycle)
	IEC 60068-2-68	Environmental testing - Part 2-68: Tests - Test L: Dust and sand
	IEC 60068-2-78	Environmental testing - Part 2-78: Tests - Test Cab: Damp heat, steady state
	IEC 60529	Degrees of protection provided by enclosures (IP Code)
	IEC 61000-6-2	Electromagnetic Compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
	IEC 61000-6-4	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
	IEC 62477-1	Safety Requirements for Power Electronic Converter Systems and Equipment - Part 1
	IEC 62909-1	Bi-Directional Grid Connected Power Converters - Part 1
	IEC 62909-2	Bi-Directional Grid Connected Power Converters - Part 2
	IEEE 519	Standard for Harmonic Control in Electric Power Systems.
	IEEE 1547	Standard for Interconnecting Distributed Resources with Electric Power Systems.
	IEEE 1547.1	Standard Conformance Test Procedures for Equipment Interconnecting Distributed Energy Resources with Electric Power Systems and Associated Interfaces.
	UL 1741	Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources.
DC Panel	IEC 61439-1	Low-voltage switchgear and controlgear assemblies - Part 1: General rules
	IEC 61439-2	Low-voltage switchgear and controlgear assemblies - Part 2: Power switchgear and controlgear assemblies
	IEC 62208	Empty enclosures for low-voltage switchgear and controlgear assemblies - General requirements
BMS	IEC 60730	Automatic electrical controls
Cabling	UL 2196	Fire Test for Circuit Integrity of Fire-Resistive Power, Instrumentation, Control and Data Cables

2.18. Design Documents

As a minimum, the design documentation listed below shall be provided. Any equipment, system, or component must be provided with its applicable design documentation, even if not explicitly listed below. Each of the design documents mentioned shall be submitted either independently or as part of a package with other documents.

Table 4: Battery Enclosure Design Documents

Component	Document	Remarks
Battery Cell	Datasheets	-

	Project Name	Doc. No.:				
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Component	Document	Remarks
	Capacity (Ah) - Voltage (V) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.
	Capacity (Ah) - Temperature (°C) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.
	SoC (%) - Voltage (V) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.
	SoH (%) - Cycles (#) curves	10,000 cycles at nominal P-Rate (0.125) and 25 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at nominal P-Rate (0.125) and 45 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at maximum P-Rate (0.25) and 25 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at maximum P-Rate (0.25) and 45 °C ambient temperature, testing in accordance with GB/T 36276.
	Overcharge curves	Voltage (V) - Time (s) curve. Current (A) - Time (s) curve. Without fire nor explosions.
	Overdischarge curves	Voltage (V) - Time (s) curve. Current (A) - Time (s) curve. Without fire nor explosions.
Battery Pack	Datasheets	-
	Capacity (Ah) - Voltage (V) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.
	Capacity (Ah) - Temperature (°C) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.

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Component	Document	Remarks
	SoC (%) - Voltage (V) curve	At nominal P-Rate (0.125) and 25 °C ambient temperature. At maximum P-Rate (0.25) and 25 °C ambient temperature.
	SoH (%) - Cycles (#) curves	10,000 cycles at nominal P-Rate (0.125) and 25 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at nominal P-Rate (0.125) and 45 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at maximum P-Rate (0.25) and 25 °C ambient temperature, testing in accordance with GB/T 36276. 10,000 cycles at maximum P-Rate (0.25) and 45 °C ambient temperature, testing in accordance with GB/T 36276.
	Overcharge curves	Voltage (V) - Time (s) curve. Current (A) - Time (s) curve. Without fire nor explosions.
	Overdischarge curves	Voltage (V) - Time (s) curve. Current (A) - Time (s) curve. Without fire nor explosions.
Battery Rack	Datasheets	Including electrical and mechanical parameters.
Battery Enclosure	Datasheets	Including power cables, communication cables, connectors, fire resistant enclosure, deflagration venting panel, drain valves, drain chute, drain connection point and paint shade.
	Short circuit current calculation report	Including simulation of faults at each characteristic point.
	Earthing sizing report	-
	Fire resistance test report	As per EN 1363-1 and EN 1363-2.
	Electrical protection coordination study	Including all electrical protection devices installed within the Battery Enclosures and the MV Stations.
	General arrangement drawings	Including main internal and external components, lifting arrangements and nameplate.
	Typical foundation drawings	-
	Logo imprint drawing	-

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Component	Document	Remarks
	Single line diagram	Including main circuits, auxiliary circuits and electrical protection devices.
	Wiring diagrams	Including LV AC cabling and DC cabling.
	Earthing drawings	-
	Control architecture diagram	-
String PCS	Datasheets	Including IGBT, DSP, DC terminals, DC electrical protection devices (fuses, surge protection devices and isolation switches), AC terminals, AC electrical protection devices (surge protection devices and automatic circuit breakers), power supply unit and control panel.
	DC electrical protection sizing report	Including DC electrical protection devices.
	AC electrical protection sizing report	Including AC electrical protection devices.
	Thermal design report	-
	Lightning protection sizing report	-
	Characteristic curves	Including efficiency curves, temperature derating curve, altitude derating curve, AC voltage derating curve, input DC voltage - output AC voltage curve and input AC voltage - output DC voltage curve.
	PQ diagrams	From Site's maximum ambient temperature to Site's minimum ambient temperature in steps of 5 °C.
	Operational functionality	Including all required modes and functions.
	PSCAD model	Including model's manual.
	PSSE model	Including model's manual.
	DigSilent model	Including model's manual.
	General arrangement drawing	Including main internal and external components, lifting arrangements and nameplate.
	Circuit diagram	-
	Wiring diagrams	-
DC Panel	Datasheets	Including busbar and electrical protection devices (fuses, BMS operated DC contactors and surge protection devices)
	Electrical protection sizing report	Including electrical protection devices.

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Component	Document	Remarks
	Busbar sizing report	-
	General arrangement drawings	-
	Single line diagram	Including electrical protection devices.
	Wiring diagrams	-
Auxiliary Panel	Datasheets	Including busbar, electrical protection devices, auxiliary transformers.
	Electrical protection sizing report	Including electrical protection devices.
	Busbar sizing report	-
	General arrangement drawings	-
	Single line diagram	Including electrical protection devices.
	Wiring diagrams	-
Thermal Management System	HVAC (electrical cabinet) datasheets	Including air temperature sensors, air relative humidity sensors and self-cleaning air filters.
	Liquid cooling/heating system datasheets	Including refrigerant, coolant, compressor, pumps, evaporator, heat exchanger, temperature meters, pressure meters, flow meters, drain pots/vessels, air vents, piping, valves, fittings and self-cleaning air filters.
	HVAC (electrical cabinet) thermal design report	Including CFD simulations.
	Liquid cooling/heating system thermal design report	Including CFD simulations.
	Liquid cooling/heating system P&ID	Including compressor, pumps, evaporator, heat exchanger, temperature meters, pressure meters, flow meters, drain pots/vessels, air vents, piping, valves, fittings and self-cleaning air filters.
	Liquid cooling/heating system isometric drawing	Including compressor, pumps, evaporator, heat exchanger, temperature meters, pressure meters, flow meters, drain pots/vessels, air vents, piping, valves, fittings and self-cleaning air filters.
	Wiring diagrams	-

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Component	Document	Remarks
Fire Detection System and Fire Suppression System	Datasheets	Including smoke detectors, heat detectors, flammable gas detectors (H2 and CO), exhaust fan, deflagration venting panel, emergency push button, fire and gas alarms (audible and visual), control panel and aerosol generators.
	General arrangement drawings	Including installation point of all components.
	Wiring diagrams	-
BMS	Datasheets	Including all hardware components and communication protocol.
	Functional specification	Including logic diagrams and software details,.
	Signals, warning and alarms list	-
	General arrangement drawings	Including all hardware components and installation points.
	Architecture diagrams	-
	Wiring diagrams	-
UPS	Datasheets	-
	Sizing report	Including power capacity, energy capacity and loads.
	General arrangement drawings	-
	Wiring diagrams	-
Others	Installation manuals	Including all components.
	Operation and maintenance manuals	Including user manuals and all components. Special attention to SoC recalibration procedures.
	MTTF, MTTR and MTBF	MTTF, MTTR and MTBF for each component.
	Spare parts list	Including any special tools.

2.19. Transportation

In case the battery packs are to be transported installed into the Battery Enclosures, measures to prevent short circuits, battery packs self-discharge and battery packs movement due to battery racks flexing derived from forces induced during transportation shall be considered.

In case the battery packs are to be transported not installed into the Battery Enclosures, and therefore are to be installed into the Battery Enclosures on Site, the OEM requirements for battery packs storage shall be followed at all times to minimize energy capacity degradation due to ambient temperature.

In any case, Battery Enclosures shall be equipped with tilt sensors and shock impact recorders throughout the whole shipping process. The tilt sensors and shock impact recorders shall be tamper-proof and detect and record impact and mishandling of the Battery Enclosures during transportation.

Battery Enclosure shipping shall be planned in such a way that the SoC of the batteries is above the supplier's recommended minimum value when they reach the Site. As batteries will be transported charged, built-in systems to avoid overcharging during transit shall be provided.

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A detailed logistics plan shall be provided ahead of shipping of any component for approval.

3. MV STATION

3.1. Scope of Supply

MV Stations shall integrate:

Table 5: Scope of Supply for MV Stations

ID	Item	Scope	Remarks
2.01	MV Station	[●]	
2.02	PCS	[●]	
2.03	LV/MV transformer	[●]	
2.04	RMU	[●]	
2.05	Auxiliary panels	[●]	
2.06	Control unit	[●]	
2.07	UPS	[●]	
2.08	Signal cables, power cables and other accessories for installation and operation	[●]	

3.2. General Requirements

MV Stations shall:

- Be offered as a modular, fully integrated, prefabricated, tested containerized solution.
- Be newly manufactured. No used, reconditioned or salvaged equipment or material shall be allowed.
- Be of proven design and include components of proven technology with sufficient operational history.
- Be manufactured in manufacturing facilities certified according to ISO 9001 (Quality Management System) and ISO 14001 (Environmental Management System) by an internationally recognized certification authority.
- Deliver rated DC power capacity at 40 °C ambient temperature within the complete SoC range (0% to 100%).
- Have looping capability and a minimum ingress protection rating of IP54.
- Have 120/120/120 fire-rated walls installed on both sides of the LV/MV transformer (meaning that fire resistance is 120 min for structural adequacy, 120 min for integrity and 120 min for insulation). These walls shall be constructed using materials certified for the required fire resistance and meet the requirements of local fire safety regulations and NFPA standards.
- Be protected from atmospheric corrosion and sand abrasion by means of protective paint in accordance with ISO 12944, the site corrosion category, as per ISO 9223, and the protective paint durability, as per ISO 12944. The protective paint shall be of a light ISO color to minimize heat absorption from solar radiation.
- Have weather resistant labelling with the arc flash level and the PPE required to operate the equipment.
- Be provided with a manual emergency push button that can be used by operators or first responders upon detecting any safety issues. The operation of this emergency push button shall shut down the MV Stations and the related Battery Enclosures immediately.
- Include electrical protection devices, lock-in-tag-out (LOTO) system and fire extinguisher.
- Integrate power, instrumentation, control and data cables in accordance with applicable standards.

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- Integrate lighting and sockets for operation and maintenance activities.

The supplier of the MV Stations shall recommend the foundations of the MV Stations in accordance with the Site climatic conditions, the soil conditions and the seismic loads of the Site, retaining maximum wind speeds recorded in the region and thermal loads caused by expected fluctuations of materials and ambient temperatures.

3.3. PCS

The PCS shall integrate:

- Power electronics module:
 - Solid state electronic devices, including IGBT.
- DC termination module with:
 - Bus terminals for DC cables connection.
 - Fuses.
 - Overcurrent protection device.
 - Overvoltage protection device.
 - Surge protection device.
 - Isolation switch.
 - Insulation monitoring device.
 - Any other required electrical protection devices.
 - Electrical protection devices shall be installed on each polarity.
- AC termination module with:
 - Bus terminals for LV cables connection.
 - Overcurrent protection device.
 - Overvoltage protection device.
 - Reverse current protection (isolation) device.
 - Surge protection device.
 - Microcontroller based automatic circuit breakers.
 - Insulation monitoring device.
 - Any other required electrical protection devices.
- Control module with:
 - External notebook (laptop) connection port for local monitoring and operation by O&M personnel.
 - Master control board.
 - Power supply unit.
 - Fieldbus communications unit.
 - Control panel including start/stop button, LED indicators for PCS status (operation (charge, discharge and idle) or stop (faults)) and emergency push button.
- Its own cooling system.
- Temperature control and moisture content control.
- Doors interlock system preventing the doors from being opened while energized.

The PCS shall be able to perform the following electrical protection functions, which triggering shall imply the power electronics tripping and the opening of the DC and AC circuits to isolate the PCS:

- Undervoltage and overvoltage protection, DC and AC.
- Overcurrent protection, DC and AC.
- Underfrequency and overfrequency protection.
- Surge protection, DC and AC.

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- Ground fault protection.
- Overheating protection.
- Smoke detection, alarm and trip.
- Insulation protection.

The PCS shall:

- Be bi-directional and suitable to provide quick responses and required reactive power.
- Be of the same manufacturer, family product, type, technology, model, series and capacity.
- Be manufactured with the same materials (brand and model) and, where possible, in the same factory.
- Be structurally identical to the ones tested and approved as part of the third-party certificates.
- Be of proven design and include proven IGBT technology with sufficient operational history.
- Deliver its rated power capacity at 40 °C ambient temperature within the complete SoC range (0% to 100%).
- Be able to generate reactive power from AC power / energy without consuming DC power / energy.
- Include self-protective and self-diagnostic features to protect itself from damage in the event of component failure or from parameters beyond safe range. The self-protective features shall not allow the PCS to be operated in a manner that may be unsafe or damaging.
- Be programable and alterable only by personnel with appropriate clearance and access.
- Have its noise emissions limited to 75 dB(A) at 1 m distance when operated at maximum electrical and thermal loads and under any weather condition.

The PCS shall have:

- Minimum ingress protection rating of IP21 for indoor installation.
- Minimum ingress protection rating of IP65 for outdoor installation.
- Provisions to prevent moisture condensation.
- Provisions to prevent the entrance of rodents, insects or any materials into air intake and exhaust ports.
- A DC voltage range wider than the DC voltage range of the battery racks.
- A maximum DC current greater than the DC current demanded by the connected battery racks during charge and discharge operating at maximum load.
- Charge and discharge efficiency greater than 98.6% at nominal load.
- Charge and discharge efficiency greater than 98.3% at any load point.
- Output current total harmonic distortion lower than 3% at rated power output.
- Output current individual harmonics lower than 1% at rated power output.

The PCS shall provide:

- Active and reactive power control.
- Grid supporting (grid following) functionalities:
 - LVRT / HVRT.
 - $\pm 100\%$ PQ.
- Grid controlling (grid forming) functionalities:
 - Voltage control.
 - Primary and secondary frequency control.
 - Ramp rate control.
 - Virtual inertia (VSG mode).
 - High short circuit reactive current contribution capability during FRT conditions.
- Local and remote monitoring and control functionalities.

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3.4. LV/MV Transformer

The LV/MV transformer shall comply with the requirements set out in Exhibit B Annex 1B Appendix 1 (HV and Electrical Requirements). The LV/MV transformer noise emissions shall be limited to 65 dB(A) at 1 m distance when operated at maximum electrical and thermal loads and under any weather condition.

3.5. RMU

The RMU shall comply with the requirements set out in Exhibit B Annex 1B Appendix 1 (HV and Electrical Requirements).

3.6. Auxiliary Panels

The auxiliary loads within the MV Station shall be supplied from the auxiliary transformers of the BESS Plant. The auxiliary panel shall allow supply of power to the thermal management system and to other devices of the MV Station through an auxiliary external power supply. The auxiliary panels shall integrate:

- Auxiliary power supply AC cable terminals.
- Copper AC busbars.
- Overcurrent protection device.
- Surge protection device.
- Circuit breakers.
- Any other required electrical protection devices.

3.7. Control Unit

The control unit shall:

- Cover the MV Station where it is installed and the Battery Enclosures related to it.
- Communicate with upper control levels (EMS, SCADA, LDC, SCMS).
- Communicate with BMS and any field equipment of the associated Battery Enclosures and MV Stations.
- Manage the PCS to set operational conditions (current, voltage and power).
- Comply with commands and setpoints given by upper control levels.
- Include self-diagnosis, self-testing and self-restoration capabilities.
- Include back-up and reset system to enable data loss management.
- Have a minimum communication speed capability of at least 10 Gbps.
- Communicate with NTP protocol to achieve time stamp synchronization.
- Include software free of virus and malware and lifetime license for such software with no cost for:
 - new versions of the software, as the same may become available;
 - updates consisting of or including modifications or improvements and routine bug fixes to the existing features of the software including specific fixes to bugs whether notified, reported by the user or included in the new versions of the software; and
 - support by e-mail, in respect of faults, failures or other issues with the installation or proper functioning of the software.
- Interoperate with the rest of the monitoring and control systems in standardized approved protocols.

The control unit shall monitor, at least, the following data:

Table 6: Data monitored by the control unit

Equipment / System	Data	Remarks
MV Station	Operational data (electrical parameters)	Measured
	Temperature and humidity	Measured
	Metering	Measured
	Operating mode / Workstate	-
	Status and alarms	-

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Equipment / System	Data	Remarks
PCS	Operational data (electrical parameters)	Measured
	DC current	Measured
	DC voltage	Measured
	AC current	Measured
	AC voltage	Measured
	Apparent power	Measured
	Active power	Measured
	Reactive power	Measured
	Power factor	Calculated
	Frequency	Measured
	Operating mode / Workstate	-
	Status and alarms	-
Auxiliary Panel	Current	Measured
	Voltage	Measured
	Power	Measured
	Status and alarms	-
BMS	Operational data (electrical parameters)	Measured
	Metering	Measured
	Operating mode / Workstate	-
	Status and alarms	-

3.8. UPS

The UPS shall:

- Have power and an energy capacity sized for 2 h backup of essential DC and AC consumers to safely shutdown and monitor the MV Station in case of loss of auxiliary power supply:
 - Control unit.
 - Electrical protection devices.
 - Emergency lighting.
- Be able to deliver its rated output under the Site climatic conditions.
- Provide the short-circuit current necessary to ensure a selective trip of downstream electrical protection devices.
- Maintain the output voltage within $\pm 1\%$ of the rated output voltage under the following conditions:
 - $\pm 5\%$ of the rated input frequency.
 - $\pm 10\%$ of the rated input voltage.
 - $\pm 100\%$ of the rated power output.
- Be installed indoor and shall be cooled by natural or forced air ventilation.

3.9. Logo Imprint

Each MV Station must have Masdar's logo imprinted on the side of the MV Station and Masdar's "Infinitor" logo imprinted on the center of the top side of the MV Station for aerial view. No other logos shall be imprinted on the MV Stations. The logos' color value shall be C100 M22 Y0 K29. Samples of the logos can be found below:

Table 7: Samples of Masdar's logos

Masdar's Infinitor Logo	Masdar's Infinitor Logo
	

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The logo imprint must be durable and resistant to wear, fading, and damage under Site climatic conditions, including exposure to radiation, temperature fluctuations, humidity, and chemicals. The logo imprint must align with Masdar's quality benchmarks for visibility and accuracy. The logos shall be in the designated positions and sizes as per Masdar's branding guidelines.

API and EPS files and Masdar's branding guidelines will be shared before the commencement of production.

3.10. Certification

The MV Stations shall have valid certifications issued by reputable ILAC-accredited testing institutions. The samples tested for such certifications must be representative of the production units. Full test reports, including the CDF and any other relevant annexes, shall be provided along with the certificates. The CDF serves as the list of components used during the certification testing of a MV Station.

To ensure that the MV Station is certified, the components in its BOM must be covered in the CDF of the required certificates and test reports. If the BOM of the MV Station does not match the CDF, the corresponding certification testing shall be repeated to certify the BOM of the MV Station.

The certifications required for the MV Station are for the following standards in their latest editions:

Table 8: MV Station Certificates

Component	Standard	Standard Name
MV Station	IEC 60529	Degrees of protection provided by enclosures (IP Code)
	ISO 834-1	Fire-resistance tests — Elements of building construction. Part 1: General requirements
	ISO 834-8	Fire-resistance tests — Elements of building construction. Part 8: Specific requirements for non-loadbearing vertical separating elements
PCS	Grid code	Electricity Transmission Code as per the Malaysia Grid Code
	IEC 60068-2-30	Environmental testing - Part 2-30: Tests - Test Db: Damp heat, cyclic (12 h + 12 h cycle)
	IEC 60068-2-68	Environmental testing - Part 2-68: Tests - Test L: Dust and sand
	IEC 60068-2-78	Environmental testing - Part 2-78: Tests - Test Cab: Damp heat, steady state
	IEC 60529	Degrees of protection provided by enclosures (IP Code)
	IEC 61000-6-2	Electromagnetic Compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
	IEC 61000-6-4	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
	IEC 62477-1	Safety Requirements for Power Electronic Converter Systems and Equipment - Part 1
	IEC 62909-1	Bi-Directional Grid Connected Power Converters - Part 1
	IEC 62909-2	Bi-Directional Grid Connected Power Converters - Part 2
	IEEE 519	Standard for Harmonic Control in Electric Power Systems.
	IEEE 1547	Standard for Interconnecting Distributed Resources with Electric Power Systems.

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Component	Standard	Standard Name
	IEEE 1547.1	Standard Conformance Test Procedures for Equipment Interconnecting Distributed Energy Resources with Electric Power Systems and Associated Interfaces.
	UL 1741	Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources.

3.11. Design Documents

As a minimum, the design documentation listed below shall be provided. Any equipment, system, or component must be provided with its applicable design documentation, even if not explicitly listed below. Each of the design documents mentioned shall be submitted either independently or as part of a package with other documents.

Table 9: MV Station Design Documents

Component	Document	Remarks
MV Station	Datasheets	Including switchgears, panels, electrical protection devices, AC LV power cables, AC MV power cables, communication cables, connectors, auxiliary transformers, fire rated walls and paint shade.
	Short circuit current calculation report	Including simulation of faults at each characteristic point.
	Earthing sizing report	-
	LV AC cable sizing report	-
	MV cable sizing report	-
	Fire resistance test report	As per ISO 834-1 and ISO 834-8.
	General arrangement drawings	Including main internal and external components, lifting arrangements, nameplate and Masdar logo imprint.
	Typical foundation drawings	-
	Logo imprint drawing	-
	Single line diagram	-
	Wiring diagrams	Including main circuits, auxiliary circuits and electrical protection devices.
	Earthing drawings	-
	Control architecture diagram	-

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Component	Document	Remarks
PCS	Datasheets	Including IGBT, DSP, DC terminals, DC electrical protection devices (fuses, surge protection devices and isolation switches), AC terminals, AC electrical protection devices (surge protection devices and automatic circuit breakers), power supply unit and control panel.
	DC electrical protection sizing report	Including DC electrical protection devices.
	AC electrical protection sizing report	Including AC electrical protection devices.
	Thermal design report	-
	Lightning protection sizing report	-
	Characteristic curves	Including efficiency curves, temperature derating curve, altitude derating curve, AC voltage derating curve, input DC voltage - output AC voltage curve and input AC voltage - output DC voltage curve.
	PQ diagrams	From Site's maximum ambient temperature to Site's minimum ambient temperature in steps of 5 °C.
	Operational functionality	Including all required modes and functions.
	PSCAD model	Including model's manual.
	PSSE model	Including model's manual.
	DigSilent model	Including model's manual.
	General arrangement drawing	Including main internal and external components, lifting arrangements and nameplate.
	Circuit diagram	-
	Wiring diagrams	-
LV/MV Transformer	Datasheets	Including tap changer, insulation oil, bushings, WTI, OTI and Buchholz relay.
	Thermal design report	-
	Off load tap changer sizing report	-
	Load and no-load losses calculation report	-
	Characteristic curves	Including efficiency curves, temperature derating curve and inrush current curve.

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Component	Document	Remarks
	General arrangement drawing	Including main internal and external components, lifting arrangements nameplate and marshalling box.
	Oil circuit diagrams	-
	Wiring diagrams	-
RMU	Datasheets	Including busbar, relays, breakers, bushings, CTs, VTs, cable terminals and surge arresters.
	Busbar sizing report	-
	Relay communication protocol	-
	Interlocking protocol	RMU-RMU and RMU-Plant Substation MV switchgear.
	General arrangement drawings	-
	Single line diagram	-
	Wiring diagrams	-
Auxiliary Panel	Datasheets	Including busbar, electrical protection devices, auxiliary transformers.
	Electrical protection sizing report	Including electrical protection devices.
	Busbar sizing report	-
	General arrangement drawings	-
	Single line diagram	Including electrical protection devices.
	Wiring diagrams	-
Control Unit	Datasheets	Including all hardware components.
	Functional specification	Including logic diagrams, software details, cybersecurity measures.
	Signals, warning and alarms list	-
	General arrangement drawings	Including all hardware components and related installation points.
	Architecture diagrams	-
	Wiring diagrams	-
UPS	Datasheets	-
	Sizing report	Including power capacity, energy capacity, loads and cooling system.
	General arrangement drawings	-
	Wiring diagrams	-

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Component	Document	Remarks
Others	Installation manuals	Including all components.
	Operation and maintenance manuals	Including user manuals and all components.
	MTTF, MTTR and MTBF	MTTF, MTTR and MTBF for each component.
	Spare parts list	Including any special tools.

3.12. Transportation

MV Stations shall be equipped with shock impact recorders throughout the whole shipping process. The shock impact recorders shall be tamper-proof and detect and record impact and mishandling of the MV Station during transportation.

4. MONITORING AND CONTROL SYSTEM

4.1. General Requirements

The monitoring and control system shall:

- Be composed of BMS, control units, EMS and SCADA as layers, hardware and software, of the system.
- Provide real-time communication by means of Modbus TCP, OPC DA or XML DA protocols.
- Permit the local and remote, manual and automatic, operation of the BESS Plant.
- Coordinate the BESS Plant operation in compliance with the grid requirements.
- Implement a firewall for safe remote access.

The monitoring and control systems for the BESS Plant shall include the most updated versions of all software. These systems must ensure seamless interaction with other instruments, sensors, and systems. The design shall encompass hardware and software components, including power supply units, controllers, I/O modules, data interface cards, and auxiliary equipment necessary for interconnection. Modifications performed on the BESS Plant during operation shall not require shutting down any devices or interrupting services. Design and equipment selection should aim for maximum standardization.

The monitoring and control systems must guarantee all functionalities of the BESS Plant as defined in this specification. Additionally, they shall integrate effectively with RTDS® systems for hardware-in-the-loop simulations. No single failure or alarm in the monitoring and control systems shall endanger personnel or compromise electricity supply. An automatic transfer functionality must be included to prevent operational tripping, ensuring no single failure or maintenance action results in a BESS Plant shutdown.

The BESS Plant operation shall be managed from the control room by authorized personnel via dedicated and redundant workstations. These workstations must display comprehensive information related to the electrical design (schematics and real-time status) and auxiliary systems. Alarms and events shall be displayed on appropriate HMIs, along with necessary actions to support operations and maintenance activities. The billing system shall operate independently of the operational systems.

The monitoring and control systems shall operate in both local and remote modes, with the ability to switch between manual and automatic operation. These modes must be coordinated with other connected systems to ensure compatibility and seamless communication. Real-time data communication must be implemented using widely accepted protocols such as Modbus TCP/IP, IEC 104, IEC 61850, and OPC. Communication links with LDC shall be redundant and independently configured over the IEC 104 protocol. The communication between the BMS at the Battery Enclosure and the control unit at the MV Station should preferably use Modbus TCP/IP for sensor communication and IEC 61850 GOOSE for electrical IEDs. Proprietary and serial protocols are not permitted. Any license or permission required to create remote links for monitoring and control shall be provided.

Connections with external systems or remote access to the BESS Plant must incorporate firewalls and cybersecurity technologies to ensure secure communication. Redundancy in hardware and software

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communication systems is mandatory to eliminate critical failure points that could lead to energy loss or disrupt operations. All interconnection equipment, such as cable glands, plates, cable terminals, connectors, label tags, isolation devices, and auxiliary tools, shall be provided. Interlocking systems, including hardware and software logic, must be independently designed from the rest of the monitoring and control equipment. Analogue and digital signal path shall be independent.

All monitoring and control equipment, devices, and cabinets must be designed to house the required hardware and software while accommodating sufficient margins to ensure optimal operation throughout the BESS Plant's design lifetime. A design margin shall also be included during system sizing to mitigate potential performance impacts over time. The monitoring and control equipment shall be scalable for handling future extensions.

The monitoring and control systems shall achieve an annual operational availability of 99.99%. System reliability shall be ensured through redundant communication links and equipment including servers, controllers, stations, switches, and networks. Redundancy factors must be considered for critical equipment and communication networks to prevent blackouts or complete outages. Critical hardware and software elements shall be designed to expedite backup support, such as spare parts, through simple operation and maintenance procedures. For software, a hot standby redundancy criterion must be employed to ensure automatic failover to backup systems during a fault.

Reliable power supply must be maintained even in the event of auxiliary equipment control failures to minimize energy loss. All monitoring and control equipment shall be supported by a redundant power supply system (UPS). Each UPS cabinet must be capable of powering all loads for at least 2 hours. Redundant power supply shall be supported with an emergency diesel generator.

Monitoring and control equipment must comply with IEC 60529 enclosure standards, with a minimum rating of IP54 for indoor installations and IP65 for outdoor installations. For equipment located in air-conditioned rooms, enclosures must meet at least IP32. Outdoor equipment cabinets must be installed under sunshades to protect against climatic conditions and, where applicable, ventilation or air conditioning shall be provided. All independent instrumentation and control equipment must offer optimal protection against electromagnetic and radiofrequency interference. Any design limitations must be explicitly stated.

4.2. EMS

The EMS shall:

- Forecast the BESS Plant operation plan as a function of:
 - The grid active and reactive power demand.
 - The BESS Plant global PCS capacity.
 - The BESS Plant global SoC.
 - The BESS Plant global auxiliary power demand.
- Coordinate the control procedures:
 - Fast power tracking response to ensure timely and accurate output power.
 - Fast frequency regulation to adjust active power when frequency exceeds the dead band.
 - Power distribution to balance SoC status of each Battery Enclosure.
 - Enable functionalities according to the operational philosophy.
- Achieve generation demand from commands:
 - Active Power Control, as per active ramp control, apparent power limitation and PID control.
 - Reactive Power Control, as per reactive ramp control, apparent power limitation and PID control.
 - Power Factor Control.
 - Primary Frequency Response.
 - Voltage Closed-Loop Control.
 - Voltage Droop Control.
- Operate the BESS Plant according to the forecasted operation plan.
- Implement a closed-loop control in real time to allow fast command distribution.
- Have all internal safety loops aligned with SIL requirements for each loop.

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- Coordinate the BESS Plant to achieve the setpoints of LDC and grid demands.
- Include self-diagnosis, self-testing and self-restoration capabilities.
- Include back-up and reset system to enable data loss management.
- Have a minimum communication speed capability of at least 10 Gbps.
- Communicate with NTP protocol to achieve time stamp synchronization.
- Include software free of virus and malware and lifetime license for such software with no cost for:
 - new versions of the software, as the same may become available;
 - updates consisting of or including modifications or improvements and routine bug fixes to the existing features of the software including specific fixes to bugs whether notified, reported by the user or included in the new versions of the software; and
 - support by e-mail, in respect of faults, failures or other issues with the installation or proper functioning of the software.
- Interoperate with the rest of the monitoring and control systems in standardized approved protocols.

The EMS shall monitor at least the following data:

Table 10: Data monitored by EMS

Data Package	Data Information
BESS Plant	Operational data (Electrical parameters from BMS and SCADA)
Plant Substation	Operational data (Electrical parameters from SCMS)
Grid	Operational data (Electrical parameters from LDC)
Alarms	General alarms and failures
User logins	Type, local or remote Time and date Parameters update Commands

4.3. SCADA

The SCADA shall:

- Display the system status along with active alarms and relevant measurements at different levels:
 - Battery Enclosure.
 - MV Station.
 - BESS Block.
 - BESS Plant.
- Monitor, collect, store and print signals, events and alarms and record sequence of events.
- Allow changing internal parameters and programming customized alerts (e.g., over-temperature alert).
- Have the ability to reset remote alarms.
- Record with a minimum resolution of 1 min, implementing SQL query technology, the monitored data along with average, minimum and maximum values for 36 rolling months.
- Include a historian to store monitored data on servers for at least 36 months.
- Process all monitored data, monitor trends and allow for history event retrieval.
- Allow for performance analysis of current conditions and historical data.
- Produce availability reports based on the contractual availability calculation.
- Calculation and reporting of any required KPIs including contractual formulas.
- Produce custom reports for any KPIs of the BESS Plant and for any data range.
- Be open and interoperable to allow integration of third party devices and sensors.
- Permit the automatic and manual operation of the BESS Plant.

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- Allow access to the database and remote access for maintenance and backup purpose.
- Implement a multiple factor authentication login for users. Users' sessions shall be automatic close after 1 hour of inactivity.
- Implement commands warnings and confirmations.
- Include self-diagnosis, self-testing and self-restoration capabilities.
- Include back-up and reset system to enable data loss management.
- Have a minimum communication speed capability of at least 10 Gbps.
- Communicate with NTP protocol to achieve time stamp synchronization.
- Include software free of virus and malware and lifetime license for such software with no cost for:
 - new versions of the software, as the same may become available;
 - updates consisting of or including modifications or improvements and routine bug fixes to the existing features of the software including specific fixes to bugs whether notified, reported by the user or included in the new versions of the software; and
 - support by e-mail, in respect of faults, failures or other issues with the installation or proper functioning of the software.
- Interoperate with the rest of the monitoring and control systems in standardized approved protocols.

The SCADA shall monitor at least the following data:

Table 11: Data monitored by SCADA

Data Package	Data Information
BESS Plant	Operation mode / Workstate Control commands (to EMS and BMS)
SCADA	Devices and servers information User parameters and permission
Other Systems	Operational data from/to LDC Commands from LDC Operational data from/to SCMS User parameters and permissions
Lighting	Status and alarms
Alarms	General alarms and failures
User logins	Type, local or remote Time and date Parameters update Commands

4.4. Servers

The minimum requirements for each of the servers are:

- Type: industrial-grade redundant servers (main and backup) for data acquisition, processing, and storage.
- CPU: Intel® Xeon® 6 or higher with at least 16 cores.
- Memory: Minimum 128 GB DDR5 RAM to handle real-time processing and historical data storage.
- Storage: RAID 5 or above configuration.
- Operating System: Windows Server.
- Database: SQL databases for storing historical data, alarms, and events.
- High-speed network interfaces (10 Gbps or higher) for seamless communication with other systems.
- High availability setup with mirrored storage and auto-failover between main and backup servers.
- Built-in hardware-based security features to prevent unauthorized access and data breaches.
- Ability to expand resources easily to handle future growth or changes in infrastructure.

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4.5. Workstations

The minimum requirements for each of the workstations are:

- Type: operator workstations with dual monitors for monitoring and control.
- CPU: Intel® Xeon® 6 or higher with at least 16 cores to handle heavy workloads.
- GPU: High-performance Nvidia® graphic cards 8 GB GDDR6 or higher for complex visualization tasks.
- Memory: Minimum 64 GB DDR5 RAM with expandability for future needs.
- Storage: Minimum 2 TB SSD for fast data access and 10 TB 7200 rpm HDD for large data storage.
- Software: HMI/SCADA software with real-time data visualization, alarms, and reporting features.
- Display: Dual 27-inch high-resolution screens for simultaneous monitoring of multiple subsystems.
- High-speed network interfaces (10 Gbps or higher) for seamless communication with servers and systems.
- Ability to expand resources easily to handle future growth or changes in infrastructure.
- Provided with ergonomic keyboards, mice, and other input devices for prolonged use.
- Include security features: secure boot, hardware encryption and advanced firewall configurations.

4.6. Communication Interfaces

All monitoring and control systems shall communicate with each other to ensure the complete operation of the entire system. Additionally, the monitoring and control system shall be capable of sharing information with relevant external monitoring and control systems, such as SCMS and LDC.

Table 12: BESS Plant interface with Plant Substation

From	To	Data Information
EMS	SCMS	Operational data (electrical parameters) for operation. Interlock system, status and alarms.
SCMS	EMS	Operational data (electrical parameters) for control synchronization. Operational data (electrical parameters) for connection scenario.

Table 13: BESS Plant interface with LDC

From	To	Data Information
EMS	LDC	Operational data (electrical parameters) for operation. Status and alarms relevant for grid operation.
LDC	EMS	Control commands for operation, as minimum: - Remote Active/Remote Power Control - Curtailment Control - Ramp Rate Control - MVar Control - Voltage Control - Power Factor Control

4.7. Cybersecurity Requirements

The cybersecurity design shall classify all cybersecurity networks with a Security Level Target (SL-T) of 3, as defined in IEC 62443, and shall adhere to fundamental security principles, including secure-by-design, defense-in-depth, least privilege, and need-to-know. These principles collectively ensure that cybersecurity measures are integrated at every stage of development, layered protections are in place to mitigate risks, access is restricted to only those who require it, and sensitive information is safeguarded from unauthorized exposure.

To achieve comprehensive security, the system shall comply with secure management practices. These include deploying next-generation firewalls, enforcing authentication, authorization, and accounting capabilities, and implementing endpoint protection. Additional measures such as sandboxing mechanisms, secure data exchange, application whitelisting, dynamic multipoint VPNs, and Zero-Trust configurations shall be incorporated to

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strengthen security defenses and minimize attack surfaces.

The cybersecurity architecture shall incorporate a zonal approach for OT environments. The design shall define and segregate multiple OT zones into critical and non-critical areas, both physically and logically, based on their functionalities, criticality, and associated risks. Secure interactions between OT and IT networks shall be facilitated through a DMZ within the OT network, ensuring controlled and monitored communication.

The cybersecurity measures shall ensure that the operational safety of the BESS Plant is maintained even in the event of a communication failure with other systems.

5. CERTIFICATION

Battery cells, packs, racks, enclosures and PCS shall be supplied along with the set of certificates that demonstrates compliance with applicable local and international codes and standards. In particular, compliance with section 6 shall be demonstrated.

6. CODES AND STANDARDS

Latest versions, including amendments, of applicable local and international codes and standards shall be considered for the design, transportation and construction of the BESS Plant, including but not limited to the systems and subsystems covered in this section.

6.1. System Level

- EN 12094-1: Fixed Firefighting Systems. Components for Gas Extinguishing Systems.
- EN 54-2: Fire Detection and Fire Alarm Systems. Control and Indicating Equipment.
- EN 54-4: Fire Detection and Fire Alarm System.
- IEC 60695-2-11: Fire Hazard Testing - Part 2-11: Glowing/Hot-Wire Based Test Methods - Glow-Wire Flammability Test Method for End-Products (GWEPT)
- IEC 60695-2-12: Fire Hazard Testing - Part 2-12: Glowing/Hot-Wire Based Test Methods - Glow-Wire Flammability Index (GWFI) Test Method for Materials.
- IEC 60695-11-10: Fire Hazard Testing - Part 11-10: Test Flames – 50 W Horizontal and Vertical Flame Test Methods.
- IEC 60695-11-20: Fire Hazard Testing - Part 11-20: Test Flames - 500 W Flame Test Methods.
- IEC 61427-2: Secondary Cells and Batteries for Renewable Energy Storage – General Requirements and Methods of Test – Part 2: On-Grid Applications.
- IEC 62485-1: Safety Requirements for Secondary Batteries and Battery Installations - Part 1: General Safety Information.
- IEC 62485-2: Safety Requirements for Secondary Batteries and Battery Installations - Part 2: Stationary Batteries.
- IEC 62933-2-1: Electrical Energy Storage (EES) Systems - Part 2-1: Unit Parameters and Testing Methods - General Specification.
- IEC 62933-2-2: Electrical Energy Storage (EES) Systems - Part 2-2: Unit Parameters and Testing Methods - Application and Performance Testing.
- IEC 62933-3-1: Electrical Energy Storage (EES) Systems - Part 3-1: Planning and Performance Assessment of Electrical Energy Storage Systems - General Specification.
- IEC 62933-3-2: Electrical Energy Storage (EES) Systems - Part 3-2: Planning and Performance Assessment of Electrical Energy Storage Systems - Additional Requirements for Power Intensive and Renewable Energy Sources Integration Related Applications.
- IEC 62933-3-3: Electrical Energy Storage (EES) Systems - Part 3-3: Planning and Performance Assessment of Electrical Energy Storage Systems - Additional Requirements for Energy Intensive and Backup Power Applications.

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- IEC 62933-5-1: Electrical Energy Storage (EES) Systems - Part 5-1: Safety Considerations for Grid-Integrated EES Systems - General Specification.
- IEC 62933-5-2: Electrical Energy Storage (EES) Systems - Part 5-2: Safety Requirements for Grid-Integrated EES Systems - Electrochemical-Based Systems.
- IEEE 1375: Guide for the Protection of Stationary Battery System.
- IEEE 1547.1: Conformance Test Procedures for Equipment Interconnecting Distributed Energy Resources with Electric Power Systems and Associated Interfaces.
- IEEE 1679.1: Characterization and Evaluation of Lithium-Based Batteries in Stationary Applications.
- IEEE 2800: Interconnection and Interoperability of Inverter-Based Resources (IBRs) Interconnecting with Associated Transmission Electric Power Systems.
- ISO 9772: Cellular Plastics - Determination of Horizontal Burning Characteristics of Small Specimens subjected to a Small Flame.
- ISO 9773: Plastics - Determination of Burning Behavior of Thin Flexible Vertical Specimens in Contact with a Small-Flame Ignition Source.
- NFPA 10: Portable Fire Extinguishers.
- NFPA 13: Installation of Sprinkler Systems.
- NFPA 15: Water Spray Fixed Systems for Fire Protection.
- NFPA 20: Installation of Stationary Pumps for Fire Protection.
- NFPA 22: Water Tanks for Private Fire Protection.
- NFPA 24: Installation of Private Fire Service Mains and Their Appurtenances.
- NFPA 68: Explosion Protection by Deflagration Venting.
- NFPA 69: Explosion Prevention Systems.
- NFPA 72: National Fire Alarm and Signaling Code.
- NFPA 80: Fire Doors and Other Opening Protectives.
- NFPA 90A: Installation of Air-Conditioning and Ventilating Systems.
- NFPA 850: Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations.
- NFPA 855: Installation of Stationary Energy Storage Systems.
- NFPA 1142: Water Supplies for Suburban and Rural Firefighting.
- NFPA 2001: Standard on Clean Agent Fire Extinguishing Systems.
- UL 864: Control Units and Accessories for Fire Alarm Systems.
- UL 94: Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.
- UL 9540: Energy Storage System Requirements.
- UL 9540A: Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems.

6.2. Battery Cells and Packs

- GB/T 36276: Lithium-ion battery for electrical energy storage.
- IEC 60529: Degrees of protection provided by enclosures (IP Code).
- IEC 62485-5: Safety Requirements for Secondary Batteries and Battery Installations - Part 5: Safe Operation of Stationary Lithium-Ion Batteries.
- IEC 62619: Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Secondary Lithium Cells and Batteries for Use in Industrial Applications.
- IEC 62620: Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Secondary Lithium Cells and Batteries for Use in Industrial Applications.

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- IEC 63056: Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Secondary Lithium Cells and Batteries for use in Electrical Energy Storage Systems.
- UL 1642: Lithium Batteries.
- UL 1973: Batteries for Use in Stationary, Vehicle Auxiliary Power and Light Electric Rail (LER) Applications.
- UL 9540A: Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems.

6.3. Battery Enclosures

- EN 1363-1: Fire Resistance Tests - Part 1: General Requirements.
- EN 1363-2: Fire Resistance Tests - Part 2: Alternative and Additional Procedures.

6.4. DC Panel / LV AC Panel

- IEC 61439-1: Low-voltage switchgear and controlgear assemblies - Part 1: General rules.
- IEC 61439-2: Low-voltage switchgear and controlgear assemblies - Part 2: Power switchgear and controlgear assemblies.
- IEC 62208: Empty enclosures for low-voltage switchgear and controlgear assemblies - General requirements.

6.5. Thermal Management System

- ASHRAE 15: Safety Standard for Refrigeration Systems.
- ASTM D3306-21: Standard Specification for Glycol Base Engine Coolant for Automobile and Light-Duty Service.
- IEEE/ASHRAE 1635: Ventilation and Thermal Management of Batteries for Stationary Applications.
- IEC 62933-5-2: Electrical energy storage (EES) systems - Part 5-2: Safety requirements for grid-integrated EES systems - Electrochemical-based systems.
- ISO 5149-1 - Refrigerating systems and heat pumps - Safety and environmental requirements.
- JIS K 2234: Engine Antifreeze Coolants.
- NFPA 855: Standard for the Installation of Stationary Energy Storage Systems.

6.6. Fire Detection System

- EN 12094-1: Fixed Firefighting Systems. Components for Gas Extinguishing Systems.
- EN 54-2: Fire Detection and Fire Alarm Systems. Control and Indicating Equipment.
- EN 54-4: Fire Detection and Fire Alarm System.
- NFPA 69: Explosion Prevention Systems.
- NFPA 72: National Fire Alarm and Signaling Code.

6.7. Fire Suppression System

- ASME B31: Code for Pressure Piping.
- NFPA 2001: Standard on Clean Agent Fire Extinguishing Systems.
- NFPA 2010: Standard for Fixed Aerosol Fire-Extinguishing Systems.

6.8. BMS

- EN 50178: Electronic Equipment for use in Power Installations.
- IEC 60730-1: Automatic Electrical Controls for Household and Similar Use - Part 1: General Requirements.
- IEC 61508: Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems.
- IEC 61131-3: Programmable Controllers - Part 3: Programming Languages.

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- IEC 61850: Communication networks and systems for power utility automation.
- ISO 13849-1: Safety of Machinery - Safety-Related Parts of Control Systems - Part 1: General Principles for Design.
- ISO 13849-2: Safety of Machinery - Safety-Related Parts of Control Systems - Part 2: Validation.
- UL 991: Safety-Related Controls Employing Solid-State Devices.
- UL 1998: Standard for Software in Programmable Components.
- UL 60730: Automatic Electrical Controls for Household and Similar Use. Particular Requirements for Temperature Sensing Controls.

6.9. PCS

- Grid Code.
- IEC 60068: Environmental Testing of Electrotechnical Products.
- IEC 60204-1: Safety of Machinery – Electrical Equipment of Machines – Part 1: General Requirements.
- IEC 61000: Electromagnetic Compatibility (EMC)
- IEC 61140: Protection Against Electric Shock - Common Aspects for Installation and Equipment.
- IEC 62477-1: Safety Requirements for Power Electronic Converter Systems and Equipment - Part 1: General.
- IEC 62909-1: Bi-Directional Grid Connected Power Converters - Part 1: General Requirements.
- IEC 62909-2: Bi-Directional Grid Connected Power Converters - Part 2: Interface of GCPC and Distributed Energy Resources.
- IEEE 519: Standard for Harmonic Control in Electric Power Systems.
- IEEE 1547: Standard for Interconnecting Distributed Resources with Electric Power Systems.
- IEEE 1547.1: Standard Conformance Test Procedures for Equipment Interconnecting Distributed Energy Resources with Electric Power Systems and Associated Interfaces.
- UL 1741: Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources.

6.10. UPS

- IEC 62040-1: Uninterruptible Power Systems (UPS) - Part 1: Safety Requirements.
- IEC 62040-2: Uninterruptible Power Systems (UPS) - Part 2: Electromagnetic Compatibility (EMC) Requirements.
- IEC 62040-3: Uninterruptible Power Systems (UPS) - Part 3: Method of Specifying the Performance and Test Requirements.
- IEC 62040-4: Uninterruptible Power Systems (UPS) - Part 4: Environmental Aspects - Requirements and Reporting.
- IEC 62040-5-3: Uninterruptible Power Systems (UPS) - Part 5-3: DC Output UPS - Performance and test Requirements.
- UL 1778: Uninterruptible Power Systems.

6.11. EMS / SCADA

- EN 50178: Electronic Equipment for use in Power Installations.
- IEC 60730-1: Automatic Electrical Controls for Household and Similar Use - Part 1: General Requirements.
- IEC 61508: Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems.
- IEC 61131-3: Programmable Controllers - Part 3: Programming Languages.
- IEC 61850: Communication networks and systems for power utility automation.

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- ISO 13849-1: Safety of Machinery - Safety-Related Parts of Control Systems - Part 1: General Principles for Design.
- ISO 13849-2: Safety of Machinery - Safety-Related Parts of Control Systems - Part 2: Validation.
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- UL 1998: Standard for Software in Programmable Components.
- UL 60730: Automatic Electrical Controls for Household and Similar Use. Particular Requirements for Temperature Sensing Controls.

6.12. Transportation

- IEC 62281: Safety of primary and Secondary Lithium Cells and Batteries During Transport.
- UN 3536: Lithium Batteries Installed in Cargo Transport Unit – Lithium-Ion Batteries or Lithium Metal Batteries.
- UN 38.3: Transport of Dangerous Goods – Lithium Batteries.

6.13. General Electrical

- IEC 60364-1: Low Voltage Electrical Installations - Part 1: Fundamental Principles, Assessment of General characteristics, Definitions.
- IEC 60364-4-41: Low Voltage Electrical Installations - Part 4-41: Protection for Safety - Protection Against Electric Shock.
- IEC 60364-5-53: Low Voltage Electrical Installations - Part 5-53: Selection and Erection of Electrical Equipment - Devices for Protection for Safety, Isolation, Switching, Control and Monitoring.
- IEC 60364-6: Low Voltage Electrical Installations - Part 6: Verification.
- IEC 62053-21: Electricity Metering Equipment - Particular Requirements - Part 21: Static Meters for AC Active Energy (classes 0.5, 1 and 2)
- IEC 62586-1: Power Quality Measurement in Power Supply Systems - Part 1: Power Quality Instruments (PQI)

6.14. Cables

- IEC 60287: Electric Cables - Calculation of the Current Rating.
- IEC 60331-2: Tests for Electric Cables under Fire Conditions - Circuit Integrity - Part 2: Test Method for Fire with Shock at a Temperature of at Least 830 °C for Cables of Rated Voltage up to and Including 0,6/1,0 kV and with an Overall Diameter Not Exceeding 20 mm.
- IEC 60332-3: Tests on Electric Cables under Fire Conditions Part 3: Tests on Bunched Wires or Cables.
- IEC 60502: Power Cables with Extruded Insulation and their Accessories for Rated Voltages from 1 kV (Um = 1,2 kV) up to 30 kV (Um = 36 kV)
- IEC 62930: Electric Cables for Photovoltaic Systems with a Voltage Rating of 1,5 kV DC.
- UL 2196: Fire Test for Circuit Integrity of Fire-Resistive Power, Instrumentation, Control and Data Cables.

6.15. Switchgear

- IEC 60947-3: Low-Voltage Switchgear and Controlgear - Part 3: Switches, Disconnectors, Switch-Disconnectors and Fuse-Combination Units.

6.16. Corrosion Categorization and Protection

- ISO 9223: Corrosion of Metals and Alloy - Corrosivity of Atmospheres - Classification, Determination and Estimation.
- ISO 12944: Paints and Varnishes - Corrosion Protection of Steel Structures by Protective Paint Systems.

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6.17. Firefighting

- ISO 15589-1: Petroleum, petrochemical and natural gas industries — Cathodic protection of pipeline systems.

6.18. Security System

- EN 50131: Alarm Systems. Intrusion and Hold-Up Systems.
- EN 50132: Alarm Systems. CCTV Surveillance for Use in Security Application.
- IEC 60839-5: Alarm and Electronic Security Systems.
- IEC 60839-7: Alarm Systems.
- IEC 62676: Video Surveillance Systems for Use in Security Applications.

6.19. Cybersecurity

- ISO/IEC 27001: Information security, cybersecurity and privacy protection — Information security management systems — Requirements.
- ISO/IEC 27002: Information security, cybersecurity and privacy protection — Information security controls.
- ISO/IEC 27019: Information security, cybersecurity and privacy protection — Information security controls for the energy utility industry.
- IEC 62351: Power systems management and associated information exchange – Data and communications security.
- IEC 62443: Security for industrial automation and control systems.
- NIST SP 800-82: Guide to Operational Technology (OT) Security.

6.20. Noise Emissions

- IEC 60034-9: Rotating Electrical Machines - Noise Limits.
- IEC 60076-10: Power Transformers - Determination of Sound Levels.
- IEC 62109: Safety of Power Converters for use in Photovoltaic Power Systems.

6.21. Reliability, Availability and Maintainability

- EN 60812: Failure Modes and Effects Analysis (FMEA and FMECA).